

ICML 2024: Teaching Segment Anything to Dive

USIS10K and USIS-SAM bring foundation-model segmentation to the murky underwater world, with the first large-scale salient-instance dataset and a SAM adapted to see beneath the surface

On land, asking a model to find the most important objects in a photo and cut out their exact silhouettes is close to routine. Point a camera underwater and everything changes. Water absorbs and scatters light, colors shift, and suspended particles (marine snow) smear across the frame. That degradation trips up even state-of-the-art segmentation models, which is a problem because underwater salient instance segmentation (USIS) is a foundational first step for marine exploration and underwater robotics.

The trouble is not only in the pixels; it is in the data. The field has long lacked a large-scale dataset with pixel-level salient-instance annotations, leaving modern models with little to train on. Meanwhile, foundation models like the Segment Anything Model (SAM) show remarkable general-purpose segmentation, but SAM was trained on natural images and does not transfer cleanly underwater. It also expects a human to supply point or box prompts, which directly conflicts with the goal of finding salient objects automatically.

This ICML 2024 paper closes both gaps at once. The authors build USIS10K, the first large-scale underwater salient instance segmentation dataset (10,632 images, pixel-level annotations, 7 categories), and propose USIS-SAM, a SAM architecture reworked for the underwater domain so the model both understands murky scenes and prompts itself. The figure below sets the scene: on the same data, USIS-SAM produces noticeably fewer spurious detections, localization errors, and instance mix-ups than competing methods.

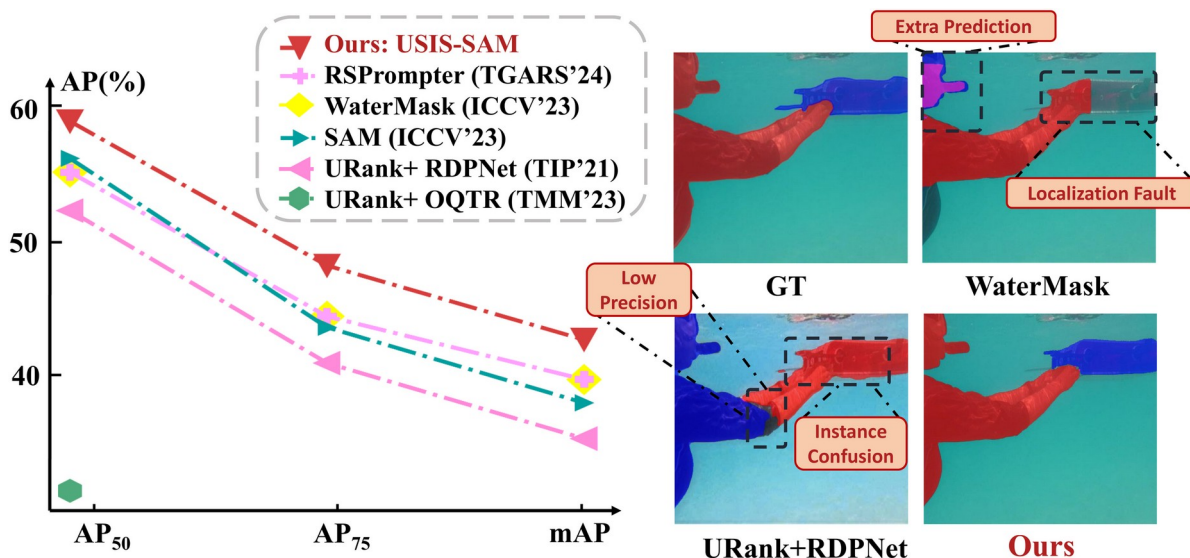


Figure 1. USIS-SAM versus recent methods on USIS10K. The line plot (left) reports AP₅₀, AP₇₅, and mAP; USIS-SAM (red) leads throughout. The panels (right) show WaterMask making extra predictions and

localization faults and URank+RDPNet showing low precision and instance confusion, while USIS-SAM stays closest to the ground truth (GT). Colors denote distinct salient instances.

Paper: <https://arxiv.org/abs/2406.06039> Code and dataset:
<https://github.com/LiamLian0727/USIS10K>

Why underwater salient segmentation is hard

Salient instance segmentation asks a model to localize the visually most important objects in an image and separate each one at the pixel level. Underwater, two difficulties compound. The first is imaging degradation: light scattering and absorption produce low contrast and color casts, and marine snow adds noise, so object boundaries blur into the background. The second is data scarcity: prior salient-instance datasets are small and mostly class-agnostic, too limited to train modern large models or to study multi-class underwater saliency in the first place.

Reaching for a foundation model like SAM is the obvious move, but the authors observe that SAM fails underwater on two fronts. Its features are not adapted to the low-contrast marine environment, and it depends on manual foreground prompts that simply do not exist in an automatic salient-segmentation setting. Their reading is that using SAM well in this domain requires two things together: a large-scale dataset that captures the diversity of underwater saliency, and an architecture that teaches SAM to see underwater and to prompt itself. The paper delivers both.

USIS10K: a large-scale underwater dataset

To break the data bottleneck, the authors assemble USIS10K: 10,632 images from a wide range of underwater scenes, each carrying pixel-level salient-instance annotations across seven categories (fish, reefs, plants, ruins, human, robot, seafloor). The examples below pair each original image (left) with its annotation mask (right), where colors mark distinct instances. The set spans clear and turbid conditions and both single-instance and crowded multi-instance scenes.

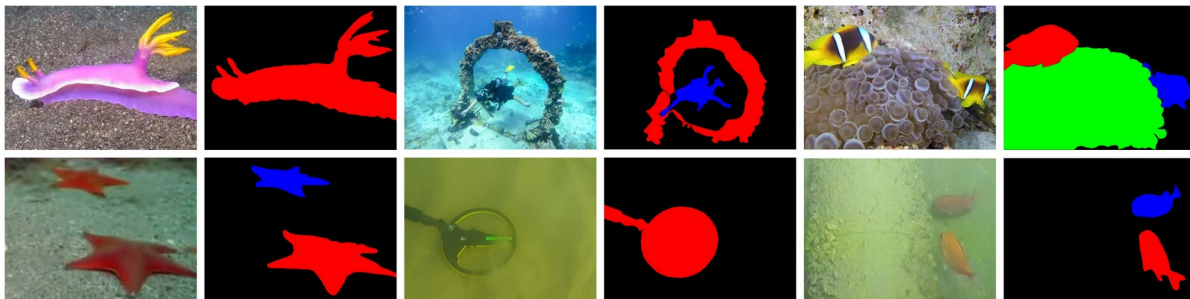


Figure 2. Annotation examples from USIS10K. Each pair shows the original underwater image (left) and its instance mask (right), with colors denoting distinct salient instances. The samples cover a range of water conditions and single- and multi-instance scenes.

The authors also analyze the dataset to characterize underwater saliency. The next figure summarizes three views: (a) the number of salient instances per category, where fish and reefs dominate; (b) the distribution of salient instances per image, compared against the land-based SIS10K; and (c) global and local color contrast for USIS10K versus SIS10K. Panel (c) is worth

pausing on: underwater imagery has systematically lower color contrast, which quantifies the intuition that the underwater setting is genuinely harder.

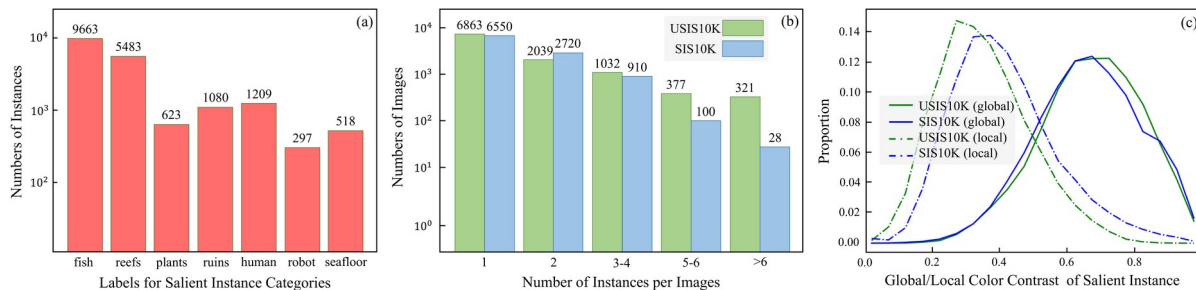


Figure 3. Key characteristics of USIS10K. (a) Salient instances per category; (b) distribution of salient instances per image, compared with SIS10K; (c) global/local color-contrast distributions for USIS10K versus SIS10K, showing lower contrast underwater.

USIS-SAM: seeing underwater and prompting itself

USIS-SAM keeps SAM's powerful pretrained backbone frozen and adds two lightweight, targeted modules. The first is the Underwater Adaptive ViT encoder (UA-ViT), which injects underwater domain knowledge into the frozen transformer. The second is the Salient Feature Prompt Generator (SFPG), which replaces SAM's manual point and box prompts by producing salient prompts automatically. The whole network trains end to end. The figure below shows the full framework alongside the internal structure of a UA-ViT block.

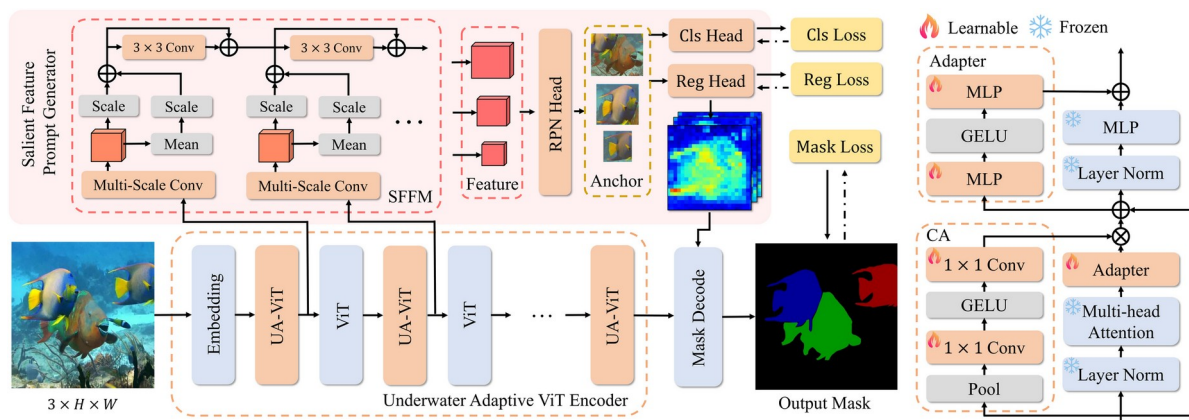


Figure 6. (a) The USIS-SAM framework: an Underwater Adaptive ViT encoder and a Salient Feature Prompt Generator (SFPG) are added on top of a frozen SAM to output instance masks end to end. (b) The UA-ViT block, where SFFM is the Salient Feature Fusion Module and CA is the Channel Adapter. Flames mark learnable parameters; snowflakes mark frozen ones.

Start with how UA-ViT learns to see underwater. It injects underwater visual cues into the frozen ViT through adapters: a channel adapter (CA) recalibrates features, and multi-scale convolutions (3×3 , 5×5 , 7×7) capture structure at different scales, balanced by an average residual that dampens noise. In short, the multi-scale feature is a sum of scaled convolutions over the channel-adapted features, then blended with its own mean by a coefficient λ before being fed back into the backbone. The strong frozen backbone gains the ability to cope with murky marine scenes while adding only a small number of parameters.

Now the prompting side. The SFPG centers on a Salient Feature Fusion Module (SFFM) that aggregates features from each UA-ViT block, generates salient prompt embeddings automatically, and passes them to SAM's mask decoder. The point is to remove the human from the loop: SFPG turns SAM from a model that waits for someone to click into one that works out of the box on salient targets. Training uses a combined loss covering region proposals (RPN), classification, box regression, and mask segmentation.

What does SFPG actually do? The figure below visualizes the features it generates. Comparing the without-SFPG and with-SFPG columns, adding SFPG concentrates the network's attention on the salient instances (the manta ray, the diver, the sculpture) while suppressing background response. This suggests SFPG does more than fuse features; it also steers attention toward the salient regions.

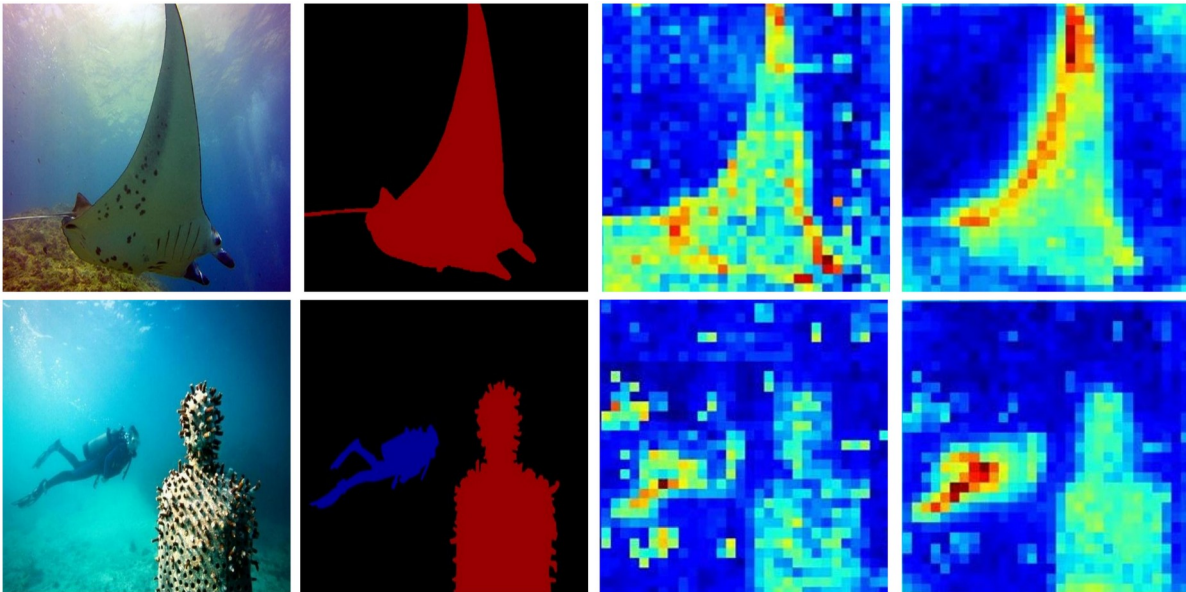


Figure 7. Visualization of features generated by the SFPG. Left to right: original image, ground truth (GT), feature response without SFPG, and with SFPG. With SFPG, the high responses (warm colors) concentrate on the salient instances while background response is suppressed.

Results: a new state of the art in both settings

The authors evaluate two settings on USIS10K. The class-agnostic setting only requires localizing and precisely masking salient objects, ignoring category; the harder multi-class setting also requires the correct class. In the class-agnostic setting, USIS-SAM reaches 59.7 mAP, ahead of the best salient method OQTR by 3.1 points, the underwater method WaterMask by 0.7, and the SAM-based RSPrompter by 1.5, with AP50 and AP75 of 81.6 and 67.7. In the harder multi-class setting, USIS-SAM scores 43.1 mAP, widening its lead over the strongest baselines to 4.4 points over WaterMask and 5.1 points over RSPrompter.

Table 1. Main results on USIS10K (mAP, %). Class-agnostic baseline scores are derived from the leads the paper reports.

Method	Class-agnostic mAP	Multi-class mAP
OQTR (TMM'23)	56.6	—

RSPrompter (TGARS'24)	58.2	38.0
WaterMask (ICCV'23)	59.0	38.7
USIS-SAM (Ours)	59.7	43.1

To check what each module contributes, the authors run ablations, reported on the harder multi-class mAP. Starting from the full model at 43.1: reverting the Underwater Adaptive ViT block to the original ViT block drops performance by 1.6 points (to 41.5), confirming that injecting underwater information through adapters genuinely helps the frozen backbone. Replacing SFPG with a generic Multi-scale Feature Enhancer costs 0.9 points (to 42.2), indicating that SFPG does more than fuse features. Additional controls, such as freezing the image encoder and simplifying the multi-scale convolution, further validate the specific design choices.

Table 2. Ablation of the key modules (multi-class mAP, %). Each module contributes a clear gain.

Configuration	Multi-class mAP
Full USIS-SAM	43.1
w/o Underwater Adaptive ViT encoder	41.5
SFPG replaced by Multi-scale Feature Enhancer	42.2

Takeaway and boundaries

The lesson here is plain: moving a powerful foundation model into a hard new domain works best when you also supply the right data and let the model prompt itself. By building USIS10K, the first large-scale underwater salient instance segmentation dataset, and by teaching SAM to see underwater through adapters while generating its own salient prompts, USIS-SAM sets a new state of the art for underwater salient instance segmentation. The authors also report that, retrained on the land-based SIS10K, the model still outperforms prior approaches, suggesting the design is not narrowly overfit to underwater imagery.

It is worth being precise about the conditions behind these numbers: a ViT-H backbone, 24 training epochs, six NVIDIA 3090 GPUs, and evaluation centered on USIS10K (with SIS10K as a generalization check). Broader, larger-scale deployment would need further validation. Encouragingly, both the dataset and the code are publicly released, giving the community a practical starting point for further work on underwater vision.